AI-Based Assessment of Power System Resilience under Climate Change-Induced Extreme Weather Events

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***Abstract* – Power system resilience analysis methods aim to assess the ability of a power system to withstand and recover from high-impact, low-probability events, such as natural disasters. These methods often entail performing AC Optimal Power Flows (OPF) using nonlinear optimization algorithms for numerous sequential Monte Carlo (MC) iterations, which require high computational resources. This may restrict the number of feasible simulations, with the risk of reducing the reliability on the analysis outcomes and limiting the scope of the resilience assessment. This paper introduces the use of Artificial Intelligence (AI) to enhance the computational performance of power system resilience analysis based on sequential MC methods. The case study demonstrates that Artificial Neural Networks (ANNs) provide an interesting solution, reducing the computational burden to a fraction of a classical OPF approaches, while keeping remarkable accuracy in the estimation of resiliency key performance indexes (KPIs) , such as energy not served (ENS) and survivability.**

Keywords— Artificial intelligence, neural network, resilience, power system, resilient planning, reXplan

I. INTRODUCTION

MC analysis is a powerful tool for assessing power system resilience. Its use has been studied extensively in recent years, and it has been shown to be an effective method for assessing the reliability and resilience of power systems [9]. MC simulations can help identify critical events that may lead to system vulnerabilities or failures. By running a sufficiently large number of simulations, analysts can observe which scenarios are most likely to result in adverse outcomes. This information is crucial for prioritizing resilience measures and developing strategies to mitigate the impact of specific events.

On the other hand, MC analysis can be quite challenging from a computational perspective. The input space for a MC analysis can be high-dimensional due to the large number of interconnected components and the variety of uncertainties. Sampling from high-dimensional spaces requires a large number of samples to adequately represent the possible states of the system, leading to increased computational requirements. Moreover, power system resilience often involves assessing rare events, such as extreme weather events or simultaneous failures of several grid components. MC simulations may require an impractical number of samples to capture these rare events accurately, making it challenging to obtain statistically meaningful results.

The above challenges are further expanded when Monte Carlo analysis is conducted together with the execution of optimal power flow (OPF) analysis. This common approach is used also in reXplan, a Python library for resilient electrical system planning under extreme hazard events [1]. Sequential MC (with or without OPF) is also proposed in several other research papers [9]. In [2], the authors assumed that all transmission lines are located in a single corridor. They use the failure probabilities of each line under different wind speed and intensity to generate MC outage scenarios. In [2], the authors assume that the failure of a tower or line creates a transmission corridor, and they use the fragility curve to generate MC outage scenarios. In [4], [5], [6], and [7], the grid is divided into different regions with different event intensities and failure probabilities. Hence, sequential MC simulation is used to generate outage scenarios for each region to determine the spatiotemporal impacts of extreme events on power systems.

The widespread use of MC methods in the analysis of power system resilience has led to the exploration of techniques that can reduce computation time, particularly in cases where complex optimizations, such as OPF, are required. In this context, the present paper proposes the use of AI, specifically ANN, as a surrogate for the majority of OPF analyses, reducing the computational time up to 90%, while keeping the calculation of the KPIs, within remarkable accuracy, compared to the OPF analysis.

A review of deep learning applications in power systems is provided in [11], where the proposed methods of analysis are mainly classified into discriminative, generative, and reinforcement learning approaches. Pandapower [13] provides a basic experiment on time-series analysis aided by ANN [14]. The commercial tool DIgSILENT PowerFactory [12] has an integrated neural network for time-series analysis. However, in all these applications the neural network is capable of approximating the load flow for only one network topology at time, based on its training data (load and generation patterns). Changes in network topology are not supported at the moment, making them not suitable for power system resilience problems, where multiple and often numerous simultaneous outages need to be evaluated.

It is important to highlight that the primary objective of this work is to leverage ANN to alleviate the computational load associated with sequential MC analysis in power system resilience applications. While, the overall description of the adopted methodology for the power system resilience and tool employed, on how these compare with other existing power system resilience assessment techniques, unique advantages and potential limitations are discussed in [1]. Therefore, this paper provides only essential information on the overall power system resiliency approach, sufficient to understand the integration of the proposed ANN method within the adopted power system resilience framework.

This paper is organized as follows: Section II presents a general overview of the proposed analysis and the resiliency analysis process under extreme hazard events. Section III describes the fully connected neural network used to analyse the electrical grid, which is illustrated in Section IV. Section V introduces the resiliency layer, which consists of fragility curves, time to repair damaged

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equipment, number of maintenance crews, and hazard event characterization. Section VI explains the analysis process, model training and model evaluation. Section VII summarizes the conclusions.

II. Analysis Overview

The reXplan python library, used in this work, employs a sequential MC analysis to simulate the impact of natural hazards on power systems. Each MC iteration consists of two phases: a stochastic outage phase and a restoration phase. In the outage phase, the hazard event intensity is either randomly or deterministically selected, and the fragility curves of power elements (e.g., overhead lines, generators, or substations) are used to determine the probability of failure of the power elements themselves. In the restoration phase, the available maintenance crews and the time-to-repair variables are used to model the recovery process of the failed elements. The same stochastic process is repeated for a sufficient number of MC iterations, to obtain reliable statistics. Fig. 1 shows a schematic diagram of the MC analysis process with an example of outage instances.

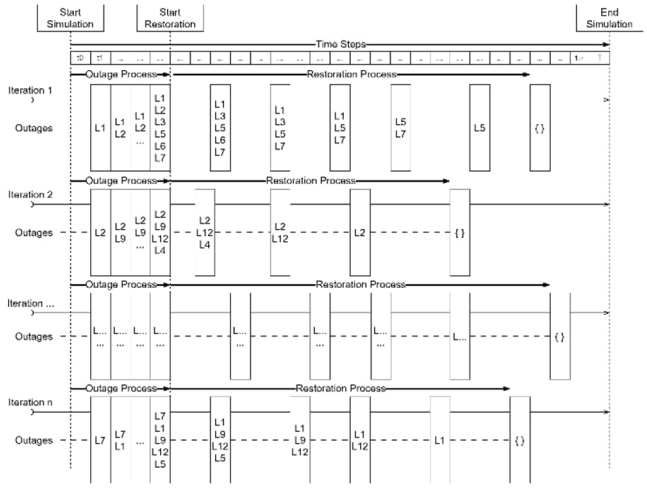


Fig. 1. Schematic diagram of sequential MC Analysis. Each iteration of the MC analysis is composed by an initial set of outages (i.e. lines), and a subsequent restoration process.

The grid topology at each time step of each MC iteration is determined by the outage and restoration processes. The grid topology, together with the operating conditions (i.e. load and generation patterns), are accounted for the OPF analysis, which calculates the loss of load for each time step. The results of the OPF analysis (i.e. loss of load) is then used to derive the KPIs. The restoration time can vary depending on the severity of the hazard events, and a large number of MC simulations are needed to obtain a reasonable confidence interval for the KPIs. The number of OPF analyses required in the study (without the use of ANN) is given by:

$$\boldsymbol{Number\ of\ OPF = T \cdot n} \quad (1)$$

Where $\boldsymbol{T}$ is the number of time steps in each MC iteration and $\boldsymbol{n}$ is the number of MC iterations. Computational effort can increase significantly if multiple scenarios are considered, for example to evaluate the impact of different mitigation measures (i.e. new network topologies). The computational burden can be reduced by using only a small fraction of the OPF analyses (i.e. 10%) to train the Artificial Neural Network (ANN). Hence, the ANN will predict the remaining 90% of the simulations. The ANN used in the paper is a fully connected neural network, as described in Section III.

III. Fully Connected Neural Network

Nowadays, the two most common neural networks used for power system analysis are the fully connected neural network (FCNN) and the family of Graph Neural Networks (GNNs) and relevant variants (i.e. Graph Convolutional Networks – GCNs).

A FCNN is a type of artificial neural network where each neuron in one layer is connected to all neurons in the next layer. Each neuron applies a linear transformation to the input vector through a weight's matrix, followed by a nonlinear activation function. Fully connected networks are "structure agnostic" i.e. they do not make any assumptions about the input. A fully connected neural network can consist of multiple layers, forming a "deep" network that can learn complex functions. On the contrary, GNN is a type of artificial neural network for processing data that can be represented as graphs. A graph is a data structure that contains nodes and edges, where nodes can be entities such as buses of an electrical grid, and edges can define the relationship between nodes. The key design element of graph neural networks is the use of pairwise message passing, such that graph nodes iteratively update their representations by exchanging information with their neighbours. GNN models can leverage a graph's connectivity information which may improves model accuracy especially for lower train dataset size. GNN can outperforms traditional FCNN in a case when the data is slightly available. However, there are several problems with GNN that need to be further studied [15].

For the above reasons, the simpler FCNN is adopted at the present stage of the research, assuming that the potential loss of accuracy is compensated by a more stable behavior of the model. The FCNN structure is schematically represented in Fig. 2.

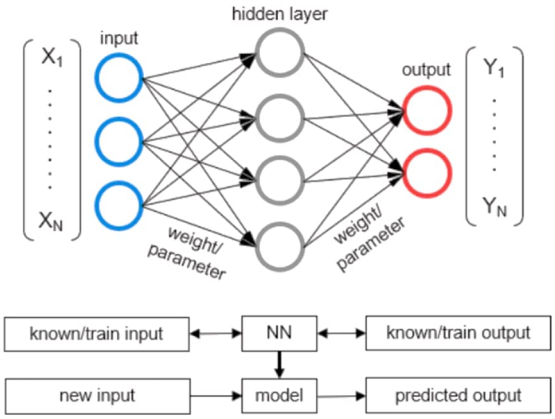


Fig. 2. Fully Connected Neural Network (source [15])

The input layer of the neural network consists of two types of variables: categorical variables and continuous variables. The former is a Boolean that indicates whether the equipment is operational or not, while the latter is a real number representing the amount of power consumed by the load, or the available capacity of the generators at a given point-in-time. The reactive power demand is highly correlated with the active power demand when the power factor is near constant, which may be the case in practical applications. To avoid collinearity problems in the neural network model, only active power is used as model feature. The categorical features (Boolean variables of the power element in-service status) are fed to the neural network without any transformation, while the continuous features are standardized to have zero mean and unit variance. This preprocessing step helps to improve the performance

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of the neural network optimization algorithm, such as gradient descent and its variants[1] [16].

The output layer is the resulting load shedding of active power necessary to keep the electrical system in operation without loading or voltage violations. From this load shedding the ENS and survivability KPIs are successively derived.

In this research, FCNN is implemented using PyTorch, a well-known framework for deep learning applications. Pytorch supports dynamic computational graphs that can change at runtime, allowing for more control and creativity over the model's structure and behaviour. It also has a high-quality GPU support that enables efficient and scalable training of large and complex models [17].

IV. The SimBench Grid

The proposed method of analysis for power system resilience is applied to a benchmark high voltage (HV) transmission system from the SimBench [10] dataset, which is a publicly available platform for power system benchmarking developed by the University of Kassel. The SimBench dataset provides electrical parameters for various power systems at different stages of development, as well as seasonal time series data for load and generation profiles at 15-minute resolution. These features enable the simulation of realistic and relevant scenarios for planning purposes. Moreover, the SimBench grid is compatible with Pandapower and the reXplan tool, which simplifies the grid modelling and analysis process.

Specifically, the selected SimBench grid is named "1-HV-urban--0-sw". The grid represents a typical 110 kV transmission system in urban areas, with the connection to the upper voltage level of 220 kV through three transformer units of 220 MVA rated power.

A summary of the power equipment included in the grid model is indicated in *TABLE I*.

TABLE I. Grid Data Overview

#	*Element*	*Quantity*
1	Bus	372
2	Load	79
3	Generator	98
4	Switch	498
5	External Grid	1
6	Line	113
7	Transformer	3
8	Measurement	1122
9	Controller	6
10	Bus Geodata	372
11	Substation	14

V. Resilience Model

The resilience model consists of an additional layer of information on top of the grid model described in Section IV. This layer includes the fragility curves of the power equipment, the hazard event intensity (or return periods, depending on the analysis type), the Time-To-Repair (TTR) of the equipment, and the number of available crews for grid restoration. ReXplan assigns failure probabilities to the power equipment based on these data and computes, with a MC approach, temporal sequences of equipment in-service status, accounting for stochastic outages and restoration processes (see Fig. 1).

The analysis conducted in this paper considers extreme wind hazard events in the range between 40 m/s to 60 m/s that may cause damage to overhead line structures. In principle, the same procedure applies for other hazard events. The input data for the resilience model is indicated in TABLE II.

TABLE II. Resilience Model Data

#	RESILIENCE MODEL DATA		
	Parameter	*Value*	*Unit*
1	TTR	6 [2]	h
2	Available Crews	5	units
3	Fragility Curve for OHL	See Fig. 3	-
4	Overhead Line Span	250	m
5	Hazard Event	40-60 m/s	(return period curves)

The analysis applies the fragility curve from Fig. 3 to all the towers in the grid model. Wind gusts with a speed between 40 m/s to 60 m/s is considered. The input is provided by return period curves, not showed in the paper.

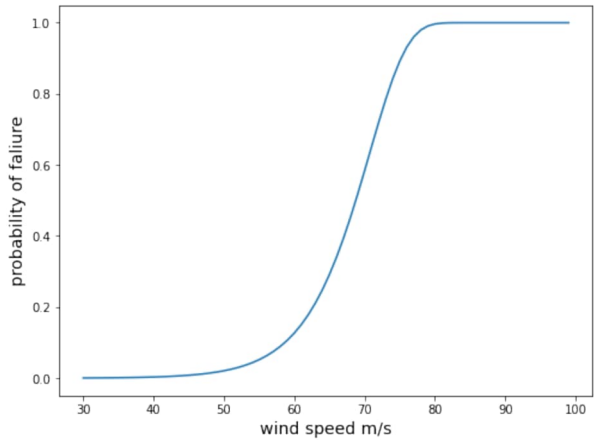


Fig. 3. Fragility curve of a transmission towers. The fragility curve is a function that provides the probability of failure based on the intensity of the event (i.e. wind speed – 3-second peak gust).

VI. AI Aided Analysis

A. Analysis Process

We demonstrate the use of ANN on two important KPIs for resiliency analysis: ENS and survivability. We calculate the KPIs for each MC iteration using OPF simulations and ANN predictions. We compare the ENS values graphically using empirical distributions, mean, and variance. We also compare the survivability curves in a diagram, showing the survival probability as a function of the minimum served load requirements.

The AI aided analysis process is composed by the following activities:

[1] Adam optimizer is used in the proposed model

[2] In the analysis, a relatively low TTR value is selected, limiting the number of points in time of the MC iterations, and the relevant computational burden for the OPF analysis.

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- Generate MC iterations of power system outages and relevant restoration process;
- Select a relatively "small" portion of grid operating scenarios (i.e. 10% of samples) for the training of the ANN – see successive explanations for the proposed methods of selecting training/test sets.
- Conduct OPF for the selected operating scenarios;
- Tune the hyperparameter of the ANN;
- Train the ANN on the complete training set;
- Evaluate the ANN performance on the test data set.

The first step is to generate MC iterations that represent the in-service status of the power elements (i.e. overhead lines). This task is part of the classical approach of reXplan. Each iteration consists of a time series of outage and restoration events, as shown in Fig. 1. The total number of operation points (or operating scenarios) is given by (1), which is the product of the number of MC iterations and the number of points in time of each MC iteration.

The second step is to build the training set. Eventually, the experiment is conducted for three different selection methods of the training data (TD):

- ANN 1: 10% of data with stratified sampling (see below explanation)
- ANN 2: 10% of data with stratified sampling + all unique combinations of outages
- ANN 3 (baseline): 10% of data without stratified sampling

The random selection, which is a common approach in machine learning (ANN 3 model) is used as baseline. On the contrary, a random stratified sampling of the training data, based on in-service status of power elements is used for ANN 1 and ANN 2 models. Specifically, for the stratified sampling, the strata are defined by grouping the MC iterations with the same initial combinations of outages. With the word 'initial' we refers to the point in time after all outages have been determined, just before the beginning of successive reparation activities. In the stratified sampling, the training set is constituted by 10% of random elements within each stratum. For the strata with only one element, we compare two approaches: (i) grouping the unique scenarios in a common stratum and then randomly sampling 10% of these elements for the model training (ANN 1), or (ii) adding all the unique scenarios to the training set (ANN 2). The first model will provide slightly better performance than the second.

The training set has been further split for 5-fold cross validation, allowing a basic hyperparameter tuning (mainly learning rate, number of epochs, hidden layers). Hyperparameter tuning has been conducted with a standard grid approach (learning rates = {0.001, 0.003, 0.01}, epochs = {300, 500, 1000}, hidden layers = {1, 4, 6}) as it was not crucial, due to the relatively simple structure of ANN models and the very good results of the analysis. Eventually, an ANN with six hidden layers, a learning rate of 0.001 and a number of 500 epochs has been used. The number of neurons per hidden layer was empirically selected equal to 512 neurons for the first four hidden layers and 256 neurons for the fifth and sixth hidden layer. No formal tuning for the number of neurons was implemented.

After hyperparameter tuning, the model has been fit on the full training data and evaluated on the test set.

B. *Artificial Neural Network Evaluation*

The evaluation of the ANN is based on the mean squared error (MSE) loss function. The MSE is calculated at each epoch for the training and test set (batch size is 256). The results are summarized in TABLE *III*.

TABLE III. Evaluation of ANN models.

ANN	**ANN Model Evaluation**		
	Parameter	***Training***	***Test***
ANN 1	MSE	$3.993 \cdot 10^{-5}$	0.030
ANN 2	MSE	$7.429 \cdot 10^{-5}$	0.027
ANN 3	MSE	$4.35 \cdot 10^{-5}$	0.033

The training loss for ANN 1 is the lowest among the three models. ANN 2 achieves the lowest test loss, indicating better generalization capabilities. ANN 3 has the higher loss on both the training and test sets, representing the lowest performance among the three models.

C. *Analysis Results*

The empirical distribution of the energy not served for the classical simulated and the ANN aided approaches are presented in Fig. 4 (ANN 1 model).

Graphically, the two distributions of Fig. 4 appear very similar. The empirical distributions for ANN 1 and ANN 3 are showed in the diagram, but a more comprehensive comparison is provided in TABLE IV (for all the three ANN models), using the mean, standard deviation, kurtosis and skewness of the two distributions. Moreover, the results of a formal Kolmogorov-Smirnov (K-S) goodness of fit are also included, to verify that the results of ANN models are statistically equivalent to the results of the OPF model. The K-S test is conducted for each ANN model vs the reference OPF model. For the OPF case, the 95% confidence interval (CI) of the mean value is indicated too.

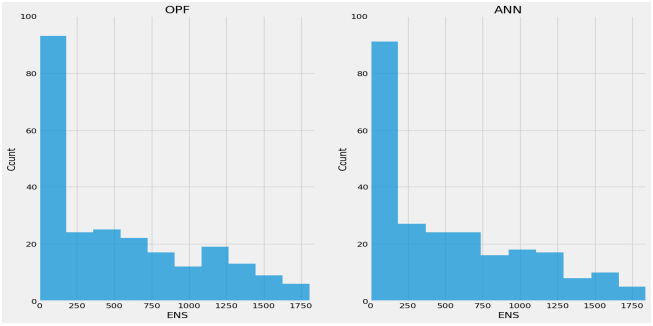


Fig. 4. Empirical density distribution of the Energy Not Served (ENS). The results of the classical simulated aproach are showed on the left subplot (OPF). The result s for ANN 1 are showed on the right subplot (ANN).

TABLE IV demonstrates the effectiveness of the ANN approach, which approximates the OPF solutions closely with a relatively low amount of training data (i.e. 10% sample). On the mean value, the error is below 1% and both mean and standard deviation for ANN models always fall within the 95% CI of the OPF model. The beneficial impact of stratified sampling on prediction accuracy is evident, as indicated by the very high p-value for the K-S statistic in the case of ANN 1, on which the stratified sampling was used.

Another KPI for resiliency analysis is survivability. Survivability measures the probability of meeting a specified

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fraction of the demand, either for the entire network or for a set of critical nodes.

TABLE IV. ENS summary. Mean, median and standard deviation values of simulated (OPF) and predicted (ANN) Energy Not Served (ENS).

ENS Summary				
Statistics	***OPF***	***ANN 1***	***ANN 2***	***ANN 3***
Mean	521.44	519.35	518.37	519.97
Mean 95% CI	[459.67, 590.77]	Within OPF 95% CI	Within OPF 95% CI	Within OPF 95% CI
Standard Devation (std)	511.43	505.14	507.84	507.62
Std 95% CI	[476.49, 552.28]	Within OPF 95% CI	Within OPF 95% CI	Within OPF 95% CI
Kurtosis	-0.588	-0.64	-0.56	-0.58
Skewness	0.766	0.74	0.77	0.77
K-S test statistic D	0	0.0333	0.0625	0.0750
K-S p value	1	0.9993	0.7377	0.5103

Survivability is computed as the ratio of MC iterations that fulfill the survival criterion to the total number of MC iterations. Fig. 5 displays the survivability values for various demand levels. The outcomes derived from ANN are quite in agreement with those from OPF computations. The values computed using the ANN fall within the 95% CI of the OPF case. There are a few exceptions, which are not practically significant, occurring primarily when survivability is very low (right-bottom corner of the diagram).

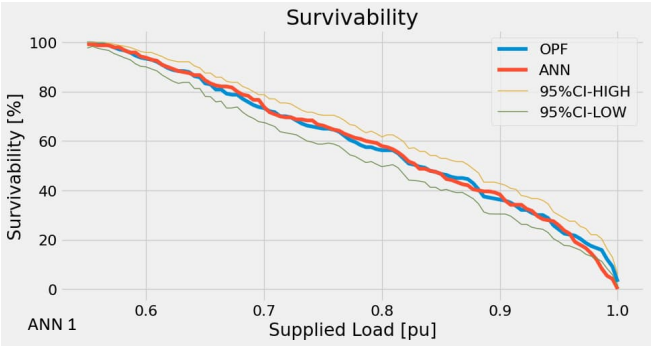


Fig. 5. Survivability curves. The diagram show the probability of success in supplying a specific portion of the maximum load, indicated on the x-axis. Results refer to ANN 1. Confidence interval refers to the OPF simulation case.

VII. Conclusions

The study demonstrates that ANN can significantly reduce computational costs associated with resiliency analysis and improve the convergence of MC simulations, compared to methods relying solely on OPF. The method has been tested on a power system resilience case using reXplan [1].

The training time for the ANN was found to be negligible in comparison to the OPF simulations. Consequently, a 90% reduction in sample size effectively translates to a 90% reduction in analysis time. Based on our project experience, simulation times for power system resilience problems can range from 8 to 24 hours when solely based on OPF, with potential extensions depending on the size of the electrical grid and the complexity of the simulation scenarios. With ANN, the computational time can be reduced to a fraction of the above range (i.e. 10% of the OPF case).

The model accuracy of ANN is demonstrated to be further enhanced by using stratified sampling of the training data, as explained in Section VI.A.

A potential direction for future research is to assess the proposed method in different resiliency scenarios, involving concurrent failures of diverse equipment types (i.e., generators, lines, transformers), or multiple hazard events.

References

[1] Pizzimbone, Luca; Jrad, Firas (2023). reXplan: A Novel Tool for the Analysis of Climate Resilience in Power Systems. TechRxiv. Preprint.

[2] Y. Yang, W. Tang, Y. Liu, Y. Xin, and Q. Wu, “Quantitative resilience assessment for power transmission systems under typhoon weather,” IEEE Access, vol. 6, pp. 40 747–40 756, 2018.

[3] M. Panteli, D. N. Trakas, P. Mancarella, and N. D. Hatziargyriou, “Boosting the power grid resilience to extreme weather events using defensiveislanding,” IEEE Transactions on Smart Grid, vol. 7, no. 6, pp. 2913–2922, Nov 2016.

[4] Espinoza, M. Panteli, P. Mancarella, and H. Rudnick, “Multi-phase assessment and adaptation of power systems resilience to natural hazards,” Electric Power Systems Research, vol. 136, pp. 352 –361, 2016.

[5] Y. Li, K. Xie, L. Wang, and Y. Xiang, “Exploiting network topology optimization and demand side management to improve bulk power system resilience under windstorms,” Electric Power Systems Research, vol. 171, pp. 127–140, 2019.

[6] M. Panteli, C. Pickering, S. Wilkinson, R. Dawson, and P. Mancarella, “Power system resilience to extreme weather: Fragility modeling, probabilistic impact assessment, and adaptation measures,” IEEE Transactions on Power Systems, vol. 32, no. 5, pp. 3747–3757, Sep. 2017.

[7] M. Panteli and P. Mancarella, “Modeling and evaluating the resilience of critical electrical power infrastructure to extreme weather events,” IEEE Systems Journal, vol. 11, no. 3, pp. 1733–1742, Sep. 2017.

[8] G. Li, Y. Huang and Z. Bie, "Reliability Evaluation of Smart Distribution Systems Considering Load Rebound Characteristics," in IEEE Transactions on Sustainable Energy, vol. 9, no. 4, pp. 1713-1721, Oct. 2018.

[9] N. Bhusal, M. Abdelmalak, M. Kamruzzaman and M. Benidris, "Power System Resilience: Current Practices, Challenges, and Future Directions," in IEEE Access, vol. 8, pp. 18064-18086, 2020.

[10] SimBench – Benchmark-Datensatz für Netzanalyse, Netzplanung und Netzbetriebsführung: https://simbench.de/de/ (last visit 05.08.2024)

[11] M. Khodayar, G. Liu, J. Wang and M. E. Khodayar, "Deep learning in power systems research: A review," in CSEE Journal of Power and Energy Systems, vol. 7, no. 2, pp. 209-220, March 2021.

[12] DIgSILENT PowerFactory. https://www.digsilent.de/en/powerfactory.html (last visit 05.08.2024)

[13] L. Thurner, A. Scheidler, F. Schäfer et al, pandapower - an Open Source Python Tool for Convenient Modeling, Analysis and Optimization of Electric Power Systems, IEEE Transactions on Power Systems.

[14] Florian Shepherd, pandapower-youtube, (2020), GitHub repository, https://github.com/FlorianShepherd/pandapower-youtube (last visit 05.08.2024)

[15] M.G.A. Fatah et Al., “Integrating Power Grid Topology in Graph Neural Network for Power Flow”, GitHub repository, gnn-powerflow/README.md at main · mukhlishga/gnn-powerflow · GitHub

[16] Daniel Voigt Godoy, “Deep Learning with Pytorch” – Volume I Fundamentals, Poland, Independently published , 2022.

[17] Nvidia web site: https://www.nvidia.com/ (last visit 05.08.2024)

[18] L. Thurner, A. Scheidler, F. Schäfer et al, pandapower - an Open Source Python Tool for Convenient Modeling, Analysis and Optimization of Electric Power Systems, IEEE Transactions on Power Systems.

[19] Coffrin, Carleton & Bent, Russell & Sundar, Kaarthik & Ng, Yeesian & Lubin, Miles. (2018). PowerModels. JL: An Open-Source Framework for Exploring Power Flow Formulations. 1-8.

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